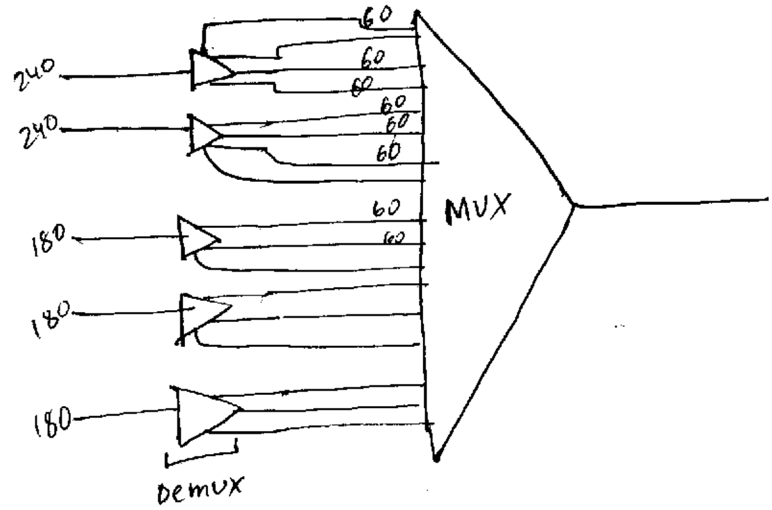
Q_1

(I) Ratio : 240 : 180 = 4 : 3

∴ Frame size = (4×2 + 3×3) + 1

= 8+9+1 = 18 bits



∴ Input bit rate = input channel rate

= 60 kbps

∴ put frame duration, T_0 = Mux unit * Input bit duration

$= 1 \times \frac{1}{60 \times 10^3}$

$= 1.667 \times 10^{-5}$ sec.

∴ Output frame rate = $\frac{1}{\text{Output frame duration}}$

$= \frac{1}{10^{-5} \times 1.667} = 60 \times 10^3$ fps

$= 60 \times 10^3$ fps ——— (II)

(III)

∴ Output frame duration = $\frac{1}{\text{output frame rate}}$

$= \frac{1}{60 \times 10^3} = 1.6667 \times 10^{-5}$ sec.

(iv) Output Data Rate = frame size × frame rate

$= 18 \times 60 \times 10^3$

$= 1080000$ bps

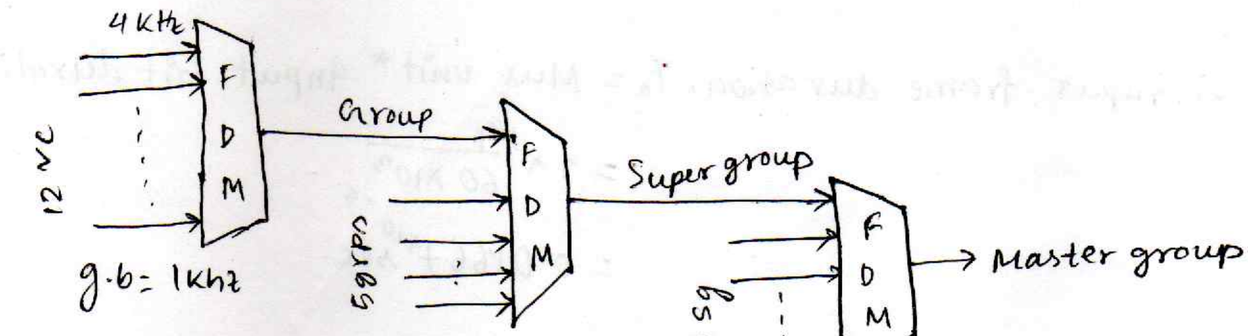
(v)

Output bit duration $= \dfrac{1}{\text{Output data rate}}$

$= \dfrac{1}{1080000} = 9.25 \times 10^{-7}$ sec.

(vi) 17 inputs

Q2



$(4 \times 12) + (11 \times 1)$
$= 59$ khz

$59 \times 5 + 4 \times 5$
$= 315$ KHz

$315 \times 10 + 10 \times 9$
$= 3240$ KHz

12 vc = 1 group

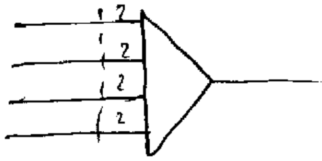
∴ 5 group = 60 vc

∴ 1 super group = 60 VC

∴ 10 super group = 600 VC = 1 mster group

Q_5

250 character/second $= 250 \times 8 = 2000$ bps



i. frame size $= 2 \times 4 + 1$

$= 9$ bits

ii. Output frame duration $= 2 \times \dfrac{1}{\text{input channel rate}}$

$= 2 \times \dfrac{1}{2000}$

$= \dfrac{1}{1000} = 1 \times 10^{-3}$ sec.

$\therefore$ Output frame rate $= \dfrac{1}{1 \times 10^{-3}}$ fps

$= 10^{3}$ fps

iii. Input bit duration $= \dfrac{1}{\text{input channel rate}}$

$= \dfrac{1}{2000} = 2 \times 10^{-3}$ sec.

$\therefore$ Input bit rate $= \dfrac{1}{2 \times 10^{-3}} = 500$ bps

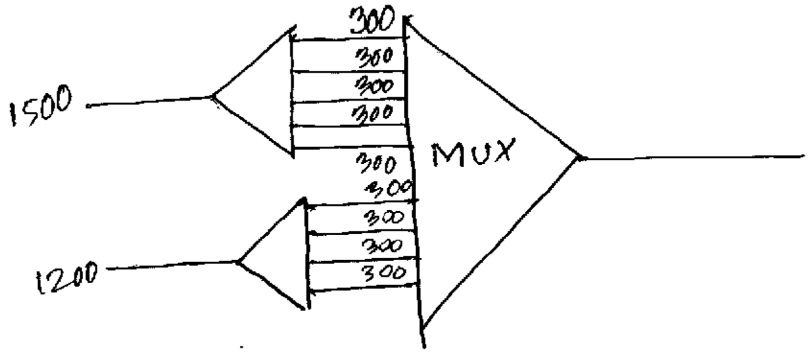
iv. Output data rate $=$ frame rate $\times$ frame size

$= 9 \times 10^{3}$ bps

v. Output slot duration $= \dfrac{\text{frame duration}}{\text{MUX unit}} = \dfrac{1 \times 10^{-3}}{2} = 5 \times 10^{-4}$ sec

$\therefore$ Output bit duration $= \dfrac{5 \times 10^{-4}}{2} = 2.5 \times 10^{-4}$ sec.

Q3



Q4

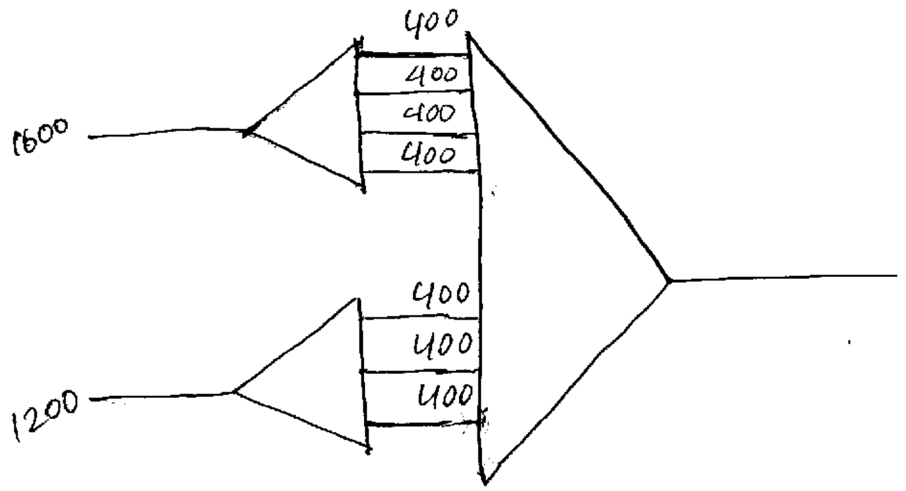
Total bandwidth $= 12 \times 5.2$
$= 62.4$ KHz

$\therefore$ Bandwidth of guardbands $= 69 - 62.4$
$= 6.6$ KHz

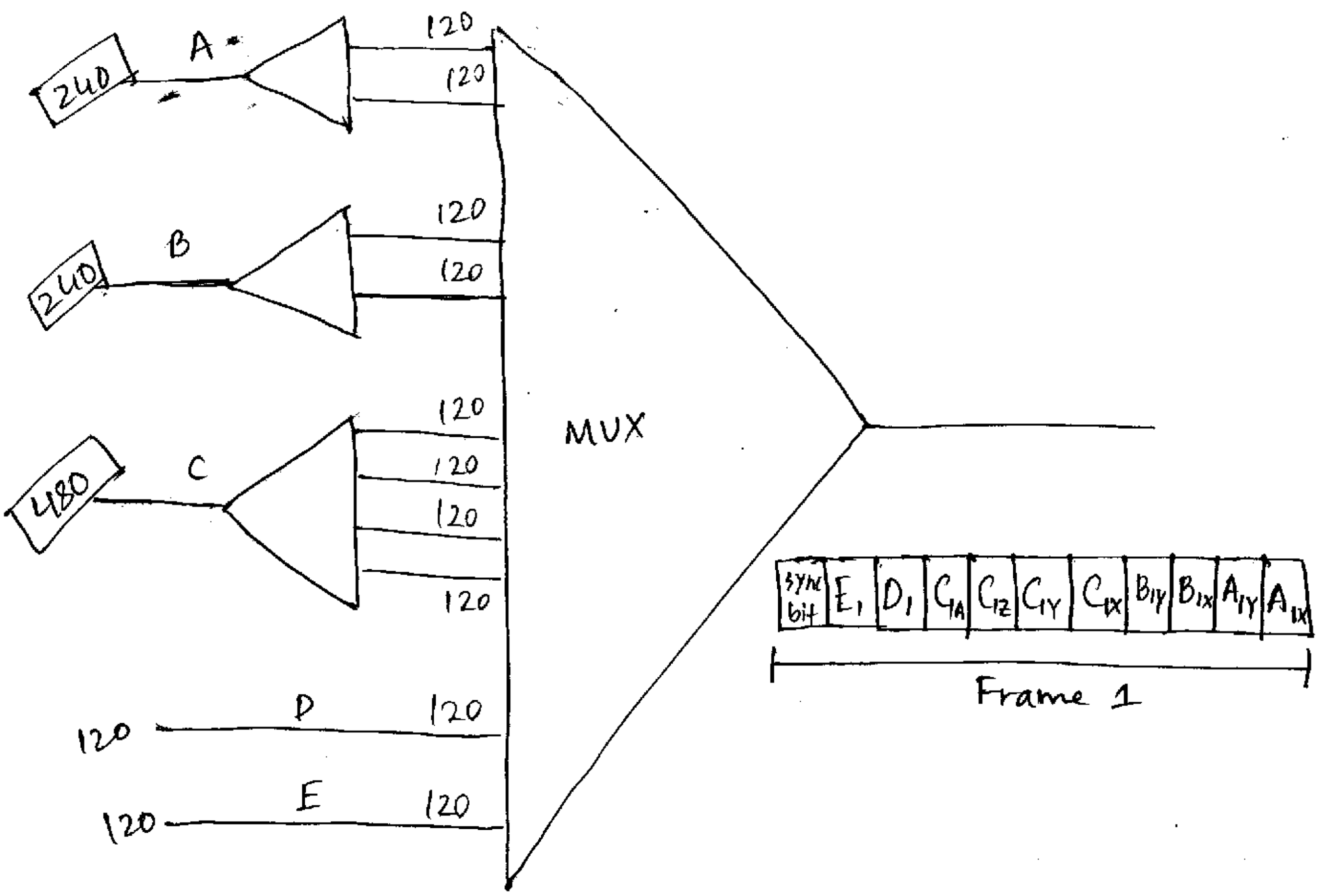
FDM needs guard bands because multiple baseband frequencies are transfered during a similar time, that have a high chance of overlapping, making the composite signal a 'জগাখিচুড়ি' that might lead to wrong output of frequencies.

TDM don't need it because at a time, one part/slot is allocated for only one time slot, resulting no overlapping.

Q6


1600
400
400
400
400
1200
400
400
400

Q7


240
A
120
120
240
B
120
120
480
C
120
120
120
120
MUX
120
D
120
120
E
120
sync bit
E1
D1
C1A
C1Z
C1Y
C1X
B1Y
B1X
A1Y
A1X
Frame 1

Q8

(i) 6×10^3 bit/sec = input data rate

6000 page/min → 300 × 6000
$= 18 \times 10^5$ bit/min
$= 30 \times 10^3$ bit/sec

5 — 30×10^3 bit/sec
1 — 6×10^3 bit/sec

(ii) input bit duration $= \frac{1}{6 \times 10^3} = 1.667 \times 10^{-4}$ sec.

(iii) Output frame rate $= \frac{1}{\text{output frame duration}}$

$= \frac{1}{\text{Mux unit} \times \text{input bit duration}}$

$= \frac{1}{2 \times 1.667 \times 10^{-4}} = 3000$ fps

(iv) frame size $= (2 \times 8 \times 5) + 1$
$= 81$ bits

(v) Output data rate $= 3000 \times 81 = 243000$ bps

(vi) Output bit duration $= \frac{1}{\text{Output data rate}} = \frac{1}{243000}$
$= 4.11 \times 10^{-6}$ sec.

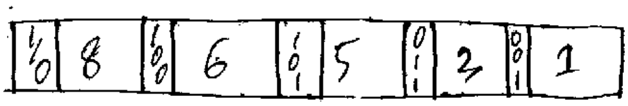
A

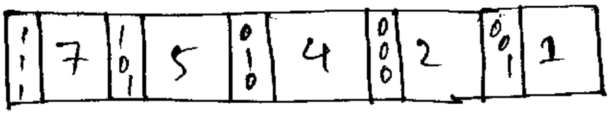
Q-9

(i) Statistical TDM

(ii)

Slot 1 ←





slot - 2

slot	over head/ addressing bits
1	001
2	000
3	011
4	010
5	101
6	100
7	111
8	110

(iii) Slot - 1

For addressing = 3 X 5 = 15 bits

Let,

MUX unit = 1 character = 8 bit

∴ For data in slots = 8 X 5 = 40 bit

∴ Total frame size = 40 + 15 = 55 bits,

Same for slot 2

Dynamic size

(iv) If the addressing bits requires more size/space than the data bits itself, it does not make any sense and it is completely reduntant, wasting spaces & bandwidths.

Q-11

guard bands $= 3 \times 3 = 9$ kHz

$\therefore$ Overall required bandwidth $= 5 + 10 + 15 + 9$
$= 39$ kHz.

Q12

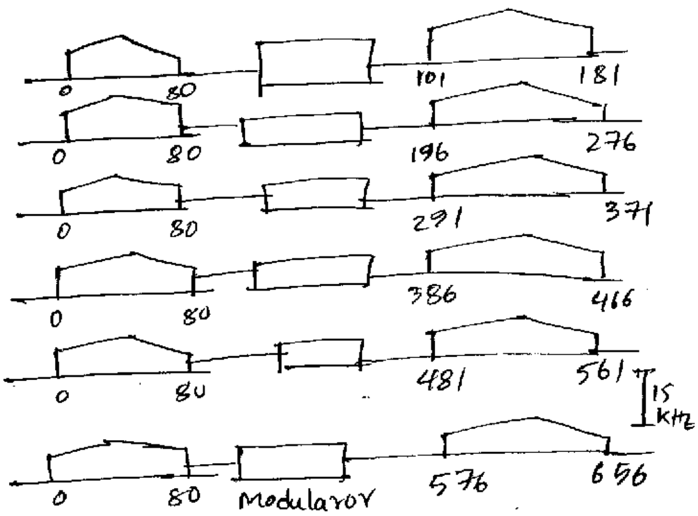
total guard bands $= 15 \times 5 = 75$ kHz

$\therefore$ Remaining overall bandwidth $= 555 - 75$
$= 480$ kHz

$\therefore$ 6 channel use 480 kHz

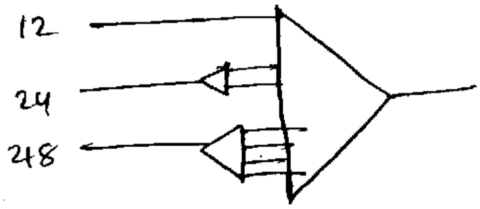
$\therefore$ 1 " " $\frac{480}{6}$ " $= 80$ kHz.

channel	data/frequency range	guard bands range
1	101 - 181	181 - 196
2	196 - 276	276 - 291
3	291 - 371	371 - 386
4	386 - 466	466 - 481
5	481 - 561	561 - 576
6	576 - 656	—



Q13

(i)



Mux unit = 4

(ii) Input bit duration = $\frac{1}{\text{input data rate}}$

$= \frac{1}{12 \times 10^3}$ bps

$= 8.33 \times 10^{-5}$ bps

(iii)

Frame size = $7 \times 4 + 1$

$= 28 + 1 = 29$ bits $= 3.625$ byte

(iv) Frame Rate = $\frac{1}{\text{frame duration}} = \frac{1}{4 \times 8.33 \times 10^{-5}}$

$= 3000.12005$ fps

(v) Output bit duration = $\frac{1}{\text{output data rate}}$

Output data rate

= frame size × frame rate

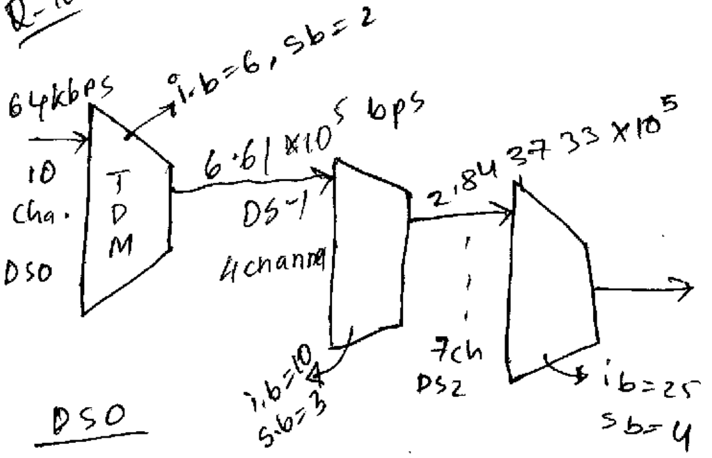
$= 3000.12005 \times 29$

$= 87003.48015$

$= \frac{1}{87003.48015}$

$= 1.1493 \times 10^{-5}$ Ans

Q-10



DSO

input channel rate $= 64\,kbps$
$= 64\times10^3\,bps$

$\therefore$ frame size $= 10\times6+2$
$= 62$ bits

Input bit duration $= \dfrac{1}{64\times10^3}$

$\therefore$ Output frame rate $= \dfrac{1}{\text{output frame duration}} = \dfrac{1}{6\times(64\times10^3)^{-1}}$
$= 10666.67$ fps

$\therefore$ Output data rate $= 62\times10666.67$
$= 661333.333$ bps

DS-1

Frame size $= 10\times4+3$
$= 43$ bits

Input bit duration $= \dfrac{1}{6.6133\times10^5}$

$\therefore$ Output frame duration $= \dfrac{10}{6.6133\times10^5}$

$\therefore$ Output frame rate $= \dfrac{6.6133\times10^5}{10}$

$\therefore$ Output data rate $=$
$43\times\dfrac{6.6133\times10^5}{10}$
$= 2.843733\times10^5$ bps

DS-2

Frame size $= 7\times25+4$
$= 179$ bits

$\therefore$ input bit duration $= \dfrac{1}{2.843733\times10^5}$

$\therefore$ Output frame duration $= 25\times\dfrac{1}{2.843733\times10^5}$

output frame rate $= \dfrac{2.843733\times10^5}{25}$ fps

$\therefore$ Data rate $=$
$179\times\dfrac{2.843733\times10^5}{25}$

$= 2036112.828$ bps

Ans